Elouann Lucas

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About Myself

Erasmus Mundus Joint Master's student in the Copernicus Master in Digital Earth (GeoData Science track) with software engineering background and experience in Earth Observation data processing and satellite imagery analysis. Specialized in algorithm development, geospatial data analysis, and operational software tools for remote sensing applications.

Work Experience

Visiting Researcher 02/03/2026 – 03/07/2026
ESA Φ-lab *Frascati, Italy*

Conducting research on satellite on-board processing systems and assessment of sensor degradation effects on image quality, directly addressing operational challenges in satellite data correction and calibration.

- Developed and implemented evaluation algorithms for assessing data quality degradation caused by sensor-induced noise and radiometric characteristics on-board the Φ-sat-2 satellite platform.
- Conducted comparative analysis of physically-based sensor degradation simulation for flood segmentation tasks; developed validation metrics to benchmark model robustness.

Application Developer – Work-Study Programme 01/09/2022 – 31/08/2024
CGI France *Larmor-Plage, France*

- Developed full-stack software features for enterprise systems
- Designed microservices architecture and wrote technical specifications for internal development frameworks focusing on scalability
- Gained experience in software development best practices, testing strategies, and continuous integration.

Education & Training

Erasmus Mundus Joint Master's – Copernicus Digital Earth 30/09/2024 – Present
Université de Bretagne Sud / Paris Lodron Universität Salzburg *Vannes, France / Salzburg, Austria*

Specialization in GeoData Science with focus on Earth Observation and satellite remote sensing. Current thesis: *Assessing the Robustness of TerraMind Foundation Model to Sensor Domain Gap for On-board Processing on the Φ-sat-2 Mission.*

- Advanced coursework: Deep Learning, Computer Vision, Machine Learning, Geospatial Data Analysis, Cloud Computing, Remote Sensing Image Processing.
- Website: master-cde.eu

Bachelor of Computer Science 01/09/2021 – 30/09/2024
Université Bretagne Sud *Vannes, France*

- Field of study: Software and applications development and analysis (180 ECTS)
- Strong foundation in algorithms, data structures, relational databases, and software architecture.
- Website: iutvannes.fr/b-u-t-informatique/

Projects & Technical Experience

Scalable Processing Pipeline for Large-Scale Satellite Imagery 31/12/2025 – 31/01/2026

Engineered end-to-end data processing pipeline for multi-source remote sensing datasets at scale.

- Designed and deployed distributed processing infrastructure for handling 100K+ satellite images from multiple optical sensors (BigEarthNet dataset).
- Implemented data preprocessing and format conversion algorithms for multi-spectral imagery; managed heterogeneous sensor data formats and radiometric standardization.
- Developed scalable data ingestion pipelines with quality assurance checkpoints and validation workflows using Python.
- Optimized data access patterns and I/O efficiency for large-scale satellite image archives.
- github.com/elucas0/big-earth-net-scalable-dl

Image Processing and Analysis for High-Resolution Sensor Data 30/09/2025 – 30/11/2025

Developed algorithms and validation pipelines for classification and analysis of high-resolution satellite and UAV imagery.

- Implemented image processing algorithms for high-resolution optical data (UAV multispectral imagery) with focus on radiometric correction and data quality assessment.
- Designed feature extraction workflows and preprocessing techniques for handling sensor modalities and characteristics.
- Developed comprehensive validation and benchmarking pipelines; conducted performance assessment against ground truth data.
- github.com/kshitijrajsharma/ml-lulc-exp

Data Management and Quality Assurance System for Environmental Monitoring 04/2022 – 06/2022

Designed and developed operational software tools for managing heterogeneous Earth Observation data and field observations.

- Architected data ingestion system for diverse sensor streams and geographic datasets with built-in validation and quality control.
- Implemented robust data pipelines with error handling, logging, and operational monitoring capabilities.
- Developed tools for data validation, statistical analysis, and quality assessment of environmental monitoring systems.
- Built user-facing software interfaces for data visualization and operational decision support.
- github.com/elucas0/SAE-PNR

Technical Skills

Remote Sensing & Earth Observation: Satellite image processing, multi-sensor data fusion, geospatial data analysis, GIS.

Programming: Python (scientific computing), algorithm implementation and validation, image processing libraries (NumPy, SciPy), data analysis workflows.

Software Development: Software architecture design, algorithm specification and documentation, CI/CD practices, Git workflow, unit and integration testing.

Data Engineering: Large-scale data processing pipelines, data format conversion and standardization, data quality assessment, sensor data handling.

Tools & Technologies: Docker, cloud computing (AWS), version control, continuous integration, relational databases.

Languages: French (Mother tongue), English (C1 – Proficient in Listening, Reading, Writing, and Speaking).

Honours and Awards

Award Winner – UBS Foundation's Call for Projects 29/03/2024
Fondation UBS

- Selected for innovative project proposal demonstrating applied problem-solving in real-world operational contexts.